

MTH:2310 DISCRETE MATH**QUIZ 6****DUE: MONDAY MAR. 31**

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic errors.

You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. (12 points) *Sequences and Summations*

Find the first five terms of the following sequences.

(a) (4 points) $\left\{ 2^n + \left\lfloor \frac{n+1}{2} \right\rfloor \right\}_{n=0}^{\infty}$

$$\begin{array}{ccccc}
 n=0 & n=1 & n=2 & n=3 & n=4 \\
 2^0 + \left\lfloor \frac{0+1}{2} \right\rfloor & 2^1 + \left\lfloor \frac{1+1}{2} \right\rfloor & 2^2 + \left\lfloor \frac{2+1}{2} \right\rfloor & 2^3 + \left\lfloor \frac{3+1}{2} \right\rfloor & 2^4 + \left\lfloor \frac{4+1}{2} \right\rfloor \\
 = 1 & = 3 & = 5 & = 10 & = 18
 \end{array}$$

(b) (4 points) $\left\{ \left(\frac{-1}{n+1} \right)^{n-2} \right\}_{n=1}^{\infty}$

$$\begin{array}{ccccc}
 n=1 & n=2 & n=3 & n=4 & n=5 \\
 \left(\frac{-1}{1+1} \right)^{1-2} & \left(\frac{-1}{2+1} \right)^{2-2} & \left(\frac{-1}{3+1} \right)^{3-2} & \left(\frac{-1}{4+1} \right)^{4-2} & \left(\frac{-1}{5+1} \right)^{5-2} \\
 = -2 & = 1 & = -\frac{1}{4} & = \frac{1}{25} & = -\frac{1}{216}
 \end{array}$$

(c) (4 points) $f_1 = -1, f_2 = 2, f_3 = 4$ and $f_{n+1} = f_n + f_{n-1} + f_{n-2}$ for $n \geq 3$

$$\begin{array}{ll}
 f_1 = -1 & f_4 = f_3 + f_2 + f_1 = 4 + 2 - 1 = 5 \\
 f_2 = 2 & f_5 = f_4 + f_3 + f_2 = 5 + 4 + 2 = 11 \\
 f_3 = 4 &
 \end{array}$$

Problem 2. (12 points) Sequences and Summations

Use an expression for the n^{th} term to show that each of the following sequences are solutions of the following recurrence relation $a_n = 3a_{n-2} + 2a_{n-3}$.

(a) (4 points) $\{2(-1)^n\}_{n=0}^{\infty}$ $a_n = 2(-1)^n$

$$\begin{aligned}
 2(-1)^n &\stackrel{?}{=} 3(2(-1)^{n-2}) + 2(2(-1)^{n-3}) \\
 &\stackrel{?}{=} 6(-1)^{n-2} + 4(-1)^{n-3} & (-1)^{n-2} &= (-1)^{n-2}(1) = (-1)^{n-2}(-1)^2 = (-1)^n \\
 &\stackrel{?}{=} 6(-1)^n - 4(-1)^n & (-1)^{n-3} &= (-1)^{n-3}(1) = (-1)^{n-3}(-1)^4 = (-1)^{n+1} \\
 & & &= -(-1)^n \\
 2(-1)^n &\stackrel{\checkmark}{=} 2(-1)^n
 \end{aligned}$$

(b) (4 points) $\{n(-1)^{n-1}\}_{n=1}^{\infty}$ $a_n = n(-1)^{n-1}$

$$\begin{aligned}
 n(-1)^{n-1} &\stackrel{?}{=} 3((n-2)(-1)^{n-3}) + 2((n-3)(-1)^{n-4}) \text{ distribute} \\
 &\stackrel{?}{=} 3n(-1)^{n-3} - 6(-1)^{n-3} + 2n(-1)^{n-4} - 6(-1)^{n-4} & (-1)^{n-3} &= (-1)^{n-3}(-1)^2 \\
 & & &= (-1)^{n-1} \\
 &\stackrel{?}{=} 3n(-1)^{n-1} - 6(-1)^{n-3} - 2n(-1)^{n-1} + 6(-1)^{n-3} & (-1)^{n-4} &= (-1)^{n-4}(-1)^4 \\
 & & &= (-1)^n = -(-1)^{n-1} \\
 n(-1)^{n-1} &\stackrel{\checkmark}{=} n(-1)^{n-1}
 \end{aligned}$$

(c) (4 points) $\{-8(2)^n\}_{n=2}^{\infty}$ $a_n = -8(2)^n$

$$\begin{aligned}
 -8(2)^n &\stackrel{?}{=} 3(-8(2)^{n-2}) + 2(-8(2)^{n-3}) \\
 &\stackrel{?}{=} -24(2)^{n-2} - 16(2)^{n-3} \\
 &\stackrel{?}{=} (2)^{n-3}(-24 \cdot 2 - 16) \\
 &\stackrel{?}{=} (2)^{n-3}(-64) \\
 &\stackrel{?}{=} (2)^{n-3} \cdot (-8)(8) & 8 &= 2^3 & 2^3 \cdot (2)^{n-3} &= 2^n \\
 -8(2)^n &\stackrel{\checkmark}{=} -8 \cdot (2)^n
 \end{aligned}$$

Problem 3. (10 points) *Recursive Definitions*

Let S be a set such that $-4 \in S$ and for any $x, y \in S$, $x + y \in S$.

(a) (6 points) Find at least 7 distinct members of the set S .

$$\boxed{-4}, \quad -4 + (-4) = \boxed{-8}, \quad -4 + (-8) = \boxed{-12}$$

$$-8 + (-8) = \boxed{-16}, \quad -8 + (-12) = \boxed{-20}$$

$$-12 + (-12) = \boxed{-24}, \quad -12 + (-16) = \boxed{-28}$$

(b) (4 points) Explicitly define S using set-builder notation.

$$S = \{x \mid (x = -4n) \wedge (n \in \mathbb{Z}^+)\}$$

Problem 4. (6 points) *Recursive Definitions*

Let $f : \mathbb{N} \rightarrow \mathbb{Q}$ be a function where $f(0) = 1$ and $f(n+1) = \left(\frac{f(n)}{f(n)+2}\right)^n$ for all $n \geq 0$.

Find $f(1)$, $f(2)$, and $f(3)$.

$$f(1) = \left(\frac{f(0)}{f(0)+2}\right)^0 = \boxed{1}$$

$$f(2) = \left(\frac{f(1)}{f(1)+2}\right)^1 = \left(\frac{1}{1+2}\right)^1 = \boxed{\frac{1}{3}}$$

$$f(3) = \left(\frac{f(2)}{f(2)+2}\right)^2 = \left(\frac{1/3}{1/3+2}\right)^2 = \left(\frac{1/3}{7/3}\right)^2 = \boxed{\frac{1}{49}}$$

Problem 5. (6 points) *Recursive Definitions*

Define a recurrence relation which produces the following sequence.

$$\{a_n\}_{n=0}^{\infty} = \{0, 1, 5, 14, 30, 55, 91, 140, \dots\}$$

$\underbrace{1}_{1} \underbrace{5}_{4} \underbrace{14}_{9} \underbrace{30}_{16} \dots$

$$a_0 = 0$$

$$a_n = a_{n-1} + n^2 \quad \text{for } n \geq 1$$

Problem 6. (14 points) *Solving Linear Recurrence Relations*Consider the recurrence relation $a_n = 2a_{n-1} + 8a_{n-2}$.

(a) (7 points) Find the general form of the solution to the recurrence relation.

char. eq'n: $r^2 = 2r + 8 \rightarrow r^2 - 2r - 8 = 0$

$$(r-4)(r+2) = 0$$

$$r = 4, -2$$

$$\boxed{r_1 = -2, r_2 = 4}$$

general form:

$$\left\{ \alpha_1 (-2)^n + \alpha_2 (4)^n \right\}_{n=0}^{\infty} \quad \text{or} \quad a_n = \alpha_1 (-2)^n + \alpha_2 (4)^n$$

(b) (7 points) Find the exact solution to the recurrence relation give the initial conditions $a_0 = 4, a_1 = 10$.

$$\begin{cases} 4 = \alpha_1 + \alpha_2 \\ 10 = -2\alpha_1 + 4\alpha_2 \end{cases} \quad 2R_1 + R_2 \rightarrow R_2 \quad \begin{cases} 4 = \alpha_1 + \alpha_2 \\ 18 = 6\alpha_2 \end{cases}$$

$$18 = 6\alpha_2 \Rightarrow \boxed{\alpha_2 = 3}$$

$$4 = \alpha_1 + \alpha_2 = \alpha_1 + 3 \Rightarrow \boxed{\alpha_1 = 1}$$

Solution

$$\left\{ (-2)^n + 3(4)^n \right\}_{n=0}^{\infty}$$